RETRAINED UNET NEURAL NETWORK TO DETECT GLAUCOMA RETINAL DISORDER WITH OCTA EYE IMAGES

A project report submitted by

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BACHELOR OF COMPUTER APPLICATIONS - DATA SCIENCE

under the guidance of

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BONAFIDE CERTIFICATE

This is to certify that the project entitled **"RETRAINED UNET NEURAL NETWORK TO DETECT GLAUCOMA RETINAL DISORDER WITH OCTA EYE IMAGES"** submitted by,

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for the award of the degree of **Bachelor of Computer Applications – Data Science** at **Amrita Vishwa Vidyapeetham, Mysuru campus** is a bonafide record of the work carried out by them under my guidance and supervision at Amrita University, Mysuru.

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hereby declare that this project report, entitled "**RETRAINED UNET NEURAL NETWORK TO DETECT GLAUCOMA RETINAL DISORDER WITH OCTA EYE IMAGES**" is a record of the original work done by us under the guidance of **Mr. Kannan M**, Assistant Professor, Department of Computer Science, Amrita Vishwa Vidyapeetham, Mysuru campus and that to the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/associate-ship/fellowship or a similar award, to any candidate in any University.

Signature of the Student

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ABSTRACT

Glaucoma is a progressive vision condition that can lead to permanent blindness. The disease can cause irreversible vision loss if left undiagnosed and untreated, affecting millions of people worldwide. Detecting glaucoma in the early stage can extend a patient's vision life. Automated disease detection is a crucial application of medical image analysis. According to a recent study, image classification algorithms based on artificial neural networks may perform more efficiently and affordably than traditional diagnostic methods. Specifically, from the clinical perspective Optical Coherence Tomography Angiography (OCTA) image analysis is emerged as a potential imaging method for glaucoma early identification and surveillance. With the abundant increase in data nowadays, traditional feature extraction algorithms are finding difficulty in coping with extracting quality features in a finite time. Also, the learning models developed from the extracted features are not so easily interpretable by humans. Therefore, considering the above-mentioned arguments in this paper we have proposed a transfer learning with U-Net for the prognosis of glaucoma disease using Retinal OCTA images. The suggested technique makes use of a convolutional neural network (CNN) known as the U-Net architecture, which is frequently employed in medical image segmentation. However, to perform the U-Net retraining operation, transfer learning mechanisms have been utilized.

1. INTRODUCTION

1.1 BROAD AREA OF RESEARCH

Globally, glaucoma, a chronic eye condition marked by increasing optic nerve damage, is a leading factor in blindness. Effective glaucoma management and therapy depend on early identification and precise diagnosis. A useful imaging method that offers high-resolution visualization of the retinal vasculature and provides insights into vascular anomalies linked to glaucoma is optical coherence tomography angiography (OCTA).

Recent years have seen a tremendous increase in interest in the use of OCTA imaging for the identification of glaucoma. Contrary to conventional imaging techniques, OCTA enables doctors to evaluate changes in retinal perfusion patterns linked to glaucomatous damage. This is because OCTA enables non-invasive visualization of the retina's microvasculature. However, interpreting OCTA pictures is a difficult, time-consuming process that calls for specialized knowledge and training.

Researchers have resorted to the use of artificial intelligence (AI) and machine learning approaches to overcome these issues and enhance the effectiveness and accuracy of glaucoma detection. In particular, utilizing Convolutional Neural Networks (CNNs), an advanced neural network design, has shown promise in automating the identification of glaucoma retinal disease using OCTA pictures.

The main goal of this field of study is to create and improve AI-based algorithms that can examine OCTA pictures to find certain biomarkers and patterns suggestive of glaucomatous alterations. These systems can learn to identify small vascular anomalies linked to glaucoma by training deep-learning models on massive datasets of annotated OCTA pictures, enabling early identification and intervention.

Data gathering and curation, algorithm creation and optimization, and validation against clinical data are a few of the major components of the research. To adequately train the deep learning models, large datasets of OCTA photos from both glaucoma patients and healthy persons are required. Experts must meticulously annotate these datasets in order to offer training and assessment models with ground truth labels.

By analyzing OCTA pictures, the proposed algorithms seek to automatically identify glaucoma retinal disorders, enabling more accurate and thorough diagnosis. The algorithms can identify small changes indicative of glaucoma development by extracting pertinent characteristics and patterns from the retinal vasculature. The algorithms can also be included in computer-aided diagnostic systems to help professionals make better choices for their patient’s care.

The ultimate objective of this study is to contribute to the creation of trustworthy and precise ML-based glaucoma diagnostic systems utilizing OCTA pictures.

These techniques can help ophthalmologists, decrease diagnostic variability, improve early detection, and offer personalized treatment approaches by automating the analytic process. Additionally, the study may offer important insights into glaucoma's underlying mechanics and enhance ocular imaging technologies.

To sum up, the whole study topic focuses on employing machine learning approaches to identify glaucoma retinal pathology using OCTA eye pictures. This study seeks to enhance the area of glaucoma care by enhancing diagnosis accuracy and enabling early intervention through the use of deep learning algorithms.

1.2 SPECIFIC AREA OF RESEARCH

Since Glaucoma can result in irreversible vision loss, this common and progressive eye illness poses a serious threat to world health. For effective treatment and care, early detection and a precise diagnosis are essential. A potential imaging method for assessing the retinal vasculature that enables non-invasive detection of glaucoma-related alterations is optical coherence tomography angiography (OCTA). However, manual OCTA picture interpretation can be laborious, subjective, and vulnerable to inter-observer variability.

Researchers have used deep learning techniques, notably convolutional neural networks (CNNs), to overcome these constraints and boost diagnostic accuracy. The U-Net is a popular CNN architecture for medical image analysis. The retraining of a UNet neural network designed for glaucoma detection in OCTA pictures can revolutionize the field of ophthalmology and improve clinical decision-making by utilizing the potential of deep learning.

The goal of this study is to retrain a UNet neural network to recognize glaucoma retinal disorders from OCTA pictures automatically. The retrained UNet model may be taught to recognize certain characteristics and patterns connected to glaucomatous alterations in the retinal vasculature by using a large dataset of annotated OCTA pictures. With the help of this automated technology, glaucoma may be accurately and reliably detected, allowing for early treatment and individualized patient care.

Transfer learning is a technique used in the retraining process that initializes the UNet model using pre-trained weights from a similar job, such as general medical picture segmentation. With this strategy, the network may utilize the knowledge amassed from enormous quantities of data and tailor it to the particular area of glaucoma diagnosis. The model's performance and accuracy may be improved by iterative training and optimization to produce cutting-edge outcomes.

The proposed study aims to accomplish a number of objectives, including the following: (1) create a reliable and effective automated system for glaucoma detection using OCTA images; (2) lessen the reliance on manual interpretation, improving efficiency and lowering diagnostic variability; (3) enable early detection and intervention, improving patient outcomes and quality of life; and (4) advance AI-assisted medical diagnostics and individualized healthcare.

In summary, the goal of this study is to retrain a UNet neural network to recognize glaucoma retinal disorders using OCTA pictures. The suggested approach has the potential to revolutionize glaucoma diagnosis, improve patient care, and progress ophthalmic research and technology by leveraging the strength of deep learning and transfer learning techniques.

1.3 BACKGROUND OF THE PROBLEM OF RESEARCH

The current gold standard for glaucoma diagnosis involves ophthalmologists manually interpreting OCTA pictures. However, this procedure requires a lot of work, takes a long time, and has inter-observer variability. Researchers are investigating the use of artificial intelligence (AI) and deep learning techniques to improve the efficiency and accuracy of glaucoma detection.

Convolutional neural networks (CNNs), in particular, have achieved outstanding results in a variety of medical imaging applications, including the processing of OCTA pictures. The UNet architecture, which was created especially for biomedical image segmentation, has demonstrated tremendous promise in the segmentation of retinal structures and the identification of disease-specific characteristics. A UNet neural network may be retrained using a sizable dataset of annotated OCTA pictures in order to create an automated system that can recognize glaucoma retinal disease.

The limits of present diagnostic techniques, which mainly rely on manual interpretation and are subject to human mistakes, are the basis of the issue. The complicated vascular architecture and modest clinical alterations found in OCTA images provide hurdles for accurate and effective glaucoma identification. By utilizing the capabilities of deep learning to automate the processing of OCTA pictures and give unbiased and accurate diagnostic information, the retraining of a UNet neural network solves these difficulties.

A pre-trained UNet model may be improved upon using a collection of annotated OCTA pictures, and the network can learn to recognize certain vascular patterns and biomarkers linked to glaucoma. Then, ophthalmologists may use this retrained UNet model as a computer-aided diagnostic tool to help with the early diagnosis and monitoring of glaucoma retinal condition. A system like this might contribute to better patient outcomes by increasing diagnostic precision, decreasing variability, and enabling more prompt treatments.

In conclusion, the backdrop issue for this research is the need for more effective and precise glaucoma detection methods employing OCTA pictures. To automate the analytic process and give impartial diagnostic help, deep learning is being used, specifically via retraining a UNet neural network. This research intends to improve early glaucoma identification and management by overcoming the shortcomings of existing diagnostic techniques, leading to better patient care and vision preservation.

1.4 OBJECTIVES OF RESEARCH

- Create a system that can automatically diagnose glaucoma: The main goal is to train a UNet neural network utilizing transfer learning methods in order to create a system that can automatically identify glaucoma retinal disorders in OCTA eye pictures. The technology should be able to reliably and effectively detect certain vascular patterns and biomarkers linked to glaucoma.
- Improved diagnostic precision: By using the capabilities of deep learning and the retrained UNet model, the research intends to improve the precision of glaucoma detection. The goal is to reduce diagnostic mistakes by training the model on a large dataset of annotated OCTA pictures, minimizing false positives and false negatives, and increasing overall accuracy.
- Reduce your reliance on manual interpretation: OCTA picture interpretation requires time and is sensitive to inter-observer variation. The project intends to decrease the reliance on manual interpretation by creating an automated system that can objectively and reliably analyze OCTA pictures. The diagnosing procedure will become more effective as a result of this goal.
- Enable early glaucoma identification and management: Timely glaucoma management and intervention depend on early glaucoma detection. The goal of the study is to use the retrained UNet model to spot modest glaucomatous alterations in the retinal vasculature, allowing for the early diagnosis of the condition. This goal will make quick treatment easier and aid in preventing additional vision loss.
- The generated system has to be tested against independent datasets and compared to the performance of human specialists in order to validate its performance. In order to achieve this goal, the retrained UNet model's sensitivity, specificity, and overall performance will be evaluated. The automated system's dependability and efficiency will be helped to assure by the validation procedure.
- Contribute to the field of AI-assisted medical diagnostics: This study seeks to make a contribution to the field of AI-assisted medical diagnostics, particularly in the field of glaucoma diagnosis. The goal is to expand the application of AI technologies in ophthalmology and stimulate the creation of intelligent diagnostic tools for better patient care by proving the effectiveness of the retrained UNet model.

The overall goal of the project is to create an accurate, automated glaucoma detection system utilizing OCTA pictures. The objectives include enhancing diagnostic precision, lowering manual interpretation, facilitating early intervention, evaluating the model's performance, and adding to the larger area of AI-assisted medical diagnostics.

1.5 APPLICATIONS AND CONTRIBUTIONS

- By autonomously analyzing OCTA eye pictures, the retrained UNet neural network can aid in the early identification of glaucoma. Early identification is essential for launching prompt management and treatment plans, which enhance patient outcomes and preserve eyesight.
- The automated system built on the retrained UNet model offers a consistent and objective method for diagnosing glaucoma. It lessens reliance on physicians' subjective judgments, minimizing inter-observer variability and improving diagnostic precision.
- A huge population may be screened for glaucoma using the retrained UNet model. The method can identify people who need additional examination and assistance by quickly analyzing OCTA pictures, maximizing healthcare resources, and minimizing pointless referrals.
- Ophthalmologists may find the automated system to be a useful decision-support tool. It helps them interpret OCTA pictures and offers more understanding and quantitative data regarding glaucomatous alterations in the retinal vasculature. Making educated clinical judgments and developing individualized treatment regimens can both benefit from this help.
- The analysis of OCTA pictures is greatly accelerated by the use of an automated system built on the retrained UNet model. It cuts down on the amount of time needed for manual interpretation, allowing ophthalmologists to devote more time to treating patients and doing other crucial duties.
- A retrained UNet neural network was developed, and its use has advanced the area of AI-assisted healthcare. It demonstrates how deep learning algorithms have the power to completely transform the medical diagnosis, especially in ophthalmology. The created system can act as a guide for future development and act as inspiration for the creation of comparable AI-based diagnostic tools in other medical specialties.
- Large datasets of annotated OCTA photos are required for the study involving the retraining of a UNet neural network. This data-driven strategy may result in the development of novel biomarkers that may further improve diagnostic precision and comprehension of the condition, as well as new insights into the pathogenesis of glaucoma.

2. LITERATURE SURVEY

2.1 GLAUCOMA DETECTION WITH OCTA DATASET

Roxane Bunod et al. [2] utilized OCTA imaging and deep learning techniques to analyze and classify different types of glaucoma, atrophic NAION, and normal control eyes. The ResNet50 CNN architecture and Integrated Gradients method were employed. The combined SCP + RPC model achieved high diagnostic performance, with area under the ROC curve (AUC) values of 0.94 for glaucoma, 0.90 for NAION, and 0.96 for normal control eyes.

Christopher Bowd et al. [6] compared the performance of a convolutional neural network (CNN) analyzing en-face vessel density images with a gradient boosting classifier (GBC) analyzing feature-based OCTA vessel density measurements and OCT retinal nerve fiber layer (RNFL) thickness measurements for classifying healthy and glaucomatous eyes. The CNN achieved significantly improved classification results compared to the GBC models, demonstrating that deep learning analysis of en-face images enhances the ability to distinguish between healthy and glaucoma eyes.

Nahida Akter et al. [7] aimed to improve the diagnostic assessment of glaucoma by analyzing new cross-sectional optic nerve head (ONH) features from optical coherence tomography (OCT) images. Deep learning (DL), and logistic regression (LR), were trained and compared for automated glaucoma detection. The DL model achieved significantly higher diagnostic performance with an accuracy of 97% on validation data and 96% on test data. A second DL model used in the pilot study produced an accuracy of 98.6% for detecting glaucoma.

Manassakorn et al. [9] introduced GlauNet, a glaucoma diagnosis network consisting of a feature-extraction section and a classification section. The network was trained and tested on a dataset of glaucomatous and non-glaucomatous eyes, achieving a sensitivity of 88.9% and specificity of 89.6%. GlauNet demonstrated robustness against artifacts and maintained sensitivity and specificity above 80% on poor-quality images. The area under the receiver operating characteristic curve for GlauNet was 0.89, indicating its effectiveness in glaucoma classification.

Atalie C. Thompson et al. [25] reviewed recent applications of deep learning models in glaucoma, highlighting their advantages and addressing the challenges associated with their development for screening, diagnosis, and progression detection. They provided an overview of deep learning compared to traditional machine learning classifiers, explored training and validation issues specific to glaucoma, and examined various scenarios where deep learning has been proposed for glaucoma, including fundus photography for screening and optical coherence tomography and standard automated perimetry for diagnosis and progression detection.

2.2 GLAUCOMA DETECTION WITH FUNDUS DATASET

Isaac Arias-Serrano et al. [1] have suggested using retinal fundus images to train MATLAB-retrained AlexNet CNN for eye disease diagnosis, notably glaucoma and diabetic retinopathy. Their technique demonstrated that re-training the AlexNet network is a strong tool for developing illness detection systems with excellent accuracy values and the ability to distinguish between more than two diseases.

Ramgopal Kashyap et al. [3] have proposed U Net deep learning model to identify and predict glaucoma before symptoms appear, where DenseNet 201 is used for feature extraction carried out by DCNN architecture and their method produced an accuracy of 98.82% and 96.90% when used for training and testing respectively.

Marriam Nawaz et al. [4] presented a glaucoma localization and classification framework. The efficientNet-B0 feature extractor was used to compute deep features from the suspected samples. Then, the Bi-directional Feature Pyramid Network (BiFPN) module of EfficientDet-D0 combined the computed features through top-down and bottom-up key points fusion. Finally, the framework predicted the localized area containing the glaucoma lesion and its associated class. The results proved that EfficientDet-D0 outperforms other frameworks.

T. Shyamalee et al. [5] have proposed a computational model for glaucoma detection in retinal fundus images utilizing data augmentation and preprocessing techniques to enhance image quality and prevent overfitting. It incorporates Inception-v3, VGG19, and ResNet50 for segmentation and classification tasks. The attention U-Net with ResNet50 achieves the highest accuracy of 99.58% in segmenting the optic disc (OD), while the Inception-v3 model achieves the highest accuracy of 98.79% for glaucoma classification.

Chayan T. I et al. [8] proposed a transfer learning model for classifying glaucoma in addition Local Interpretable Model-Agnostic Explanations (LIME) were incorporated to provide explainability, allowing medical professionals to obtain valuable information for making informed judgments and the model was able to classify Glaucoma with 94.71% accuracy.

Adnan Haider et al. [10] proposed two networks, SLS-Net and SLSR-Net, for accurate pixel-wise segmentation of the optic cup (OC) and optic disc (OD) in retinal fundus images, which maintains a large final feature map, minimizing spatial information loss and enhancing segmentation accuracy. SLSR-Net incorporates external residual connections for feature empowerment. Both networks have utilized separable convolutions for efficiency and have a low parameter count while achieving high segmentation accuracy. Evaluation of four retinal fundus image datasets demonstrated that these networks outperform existing segmentation architectures.

H.N. Veena et al. [11] proposed an efficient glaucoma framework that utilizes deep learning and a novel CNN architecture to segment the optic cup and optic disc for calculating the Cup-to-Disc Ratio (CDR). The system employed two separate CNN models to achieve more accurate results in segmenting the Optic Cup (OC) and Optic Disc (OD). The model is trained and evaluated on the publicly available DRISHTI - GS database, achieving an accuracy of 98% for optic disc segmentation and 97% for optic cup segmentation.

M.B. Sudhan et al. [12] developed a deep-learning method for an early glaucoma prediction model. It used the U-Net architecture to segment the optic cup, the DenseNet-201 pre-trained transfer learning model to extract the features, and the deep convolutional neural network (DCNN) to classify the data. The main objective was to detect glaucoma using retinal fundus images and assess whether or not a patient had glaucoma. The model's performance was evaluated using accuracy, precision, recall, specificity, and F-measure metrics. A comparative analysis was conducted with other deep learning models such as VGG-19, Inception ResNet, ResNet 152v2, and DenseNet-169. The suggested model outperformed the other models, with 98.82% accuracy during training and 96.90% accuracy during testing.

M. M. Mahdi et al. [13] provided a framework for the detection of glaucoma using retinal pictures that are based on deep learning. For image classification, the suggested method used two CNN-based models, Inception and DenseNet. To enhance training and validation procedures, transfer learning was used, and a pipeline with fewer trainable parameters was created for the intended job. The proposed model's performance achieved accuracy, precision, and recall scores of 93.84%, 92.83%, and 95.00%, respectively.

Shanmugam P et al. [14] proposed a glaucoma recognition technique based on estimating the Cup to Disc Ratio (CDR) from fundus images. The proposed technique included image acquisition, feature extraction, and glaucoma evaluation stages. The CDR ratio was computed to assess glaucoma, and a random forest classifier was used for classification based on CDR values. The proposed method achieved a classification accuracy of 99% and outperformed other techniques such as Deformable U-Net, Full-Deformable U-Net, and Original U-Net in terms of segmentation accuracy with a 14% improvement.

Mir Tanvir Islam et al. [15] developed a new glaucoma classification technique using multiple deep-learning approaches and a dataset of 634 color fundus images was annotated by eye specialists. Deep learning models such as EfficientNet, MobileNet, DenseNet, and GoogLeNet were utilized to detect glaucoma. The

EfficientNet-b3 architecture achieved the best performance with high accuracy, F1-score, and ROC AUC values of 0.9652, 0.9512, and 0.9574, respectively. Additionally, a new dataset was created by segmenting blood vessels using a U-net model, and the MobileNet v3 model achieved satisfactory results.

Amsa Shabbir et al. [16] examined various aspects of glaucoma, including its types, causes, treatment, publicly available image benchmarks, performance metrics, and detection approaches using digital image processing, computer vision, and deep learning. The article explores published research models for glaucoma detection, discussing their strengths and weaknesses. It summarizes results using tabular representations and concludes with findings and future research directions for glaucoma detection.

Law Kumar Singh et al. [17] proposed a model for predicting glaucoma in retinal fundus images where the model combines four machine learning algorithms and achieves a classification accuracy of 98.60%. In comparison, other existing models like SVM, K-nearest neighbors, and Naïve Bayes achieve accuracies of 97.61%, 90.47%, and 95.23% respectively. The results highlight the effectiveness of the proposed model for early glaucoma diagnosis in retinal fundus images.

Rutuja Shinde [18] introduced an offline Computer-Aided Diagnosis (CAD) system for glaucoma diagnosis using retinal fundus images. It utilized Le-Net for image validation, the brightest spot algorithm for ROI detection, U-Net for optic disc and optic cup segmentation, and SVM, Neural Network, and Adaboost for classification. The results show high accuracy for input image validation (99%) and ROI extraction (98.67%). Optic disc and optic cup segmentation achieved good dice coefficients (0.93, 0.87). For classification, 100% accuracy, recall, specificity, and sensitivity were achieved.

J. Raja et al. [19] presented a novel computer-supported analysis for glaucoma recognition using fundus images, where it employed a Support Vector Machine based VGG-19 network architecture. Glaucoma is identified using a CDR threshold value of 0.41, with fundus images above the threshold considered glaucoma-affected and that below-considered non-glaucoma. The proposed system achieves a classification precision of 94% for a set of 175 fundus images, demonstrating its effectiveness with commonly used digital color fundus images.

Rabbia Mahum et al. [20] proposed a deep learning-based approach for the early detection of glaucoma using retinal fundus images. It involved pre-processing of the images, followed by region of interest (ROI) extraction through segmentation. Hybrid feature descriptors, including CNN, LBP, HOG, and SURF was used to extract features of the optic disc. Employed Multi-class classifiers (SVM, RF, and KNN) for classifying images as healthy or diseased. Results demonstrated that the proposed model, using the RF algorithm with HOG, CNN, LBP, and SURF descriptors, achieves high accuracy (>99%) and 98.8% accuracy on k-fold cross-validation for early glaucoma detection.

Manar Aljazaeri et al. [21] proposed a deep learning-based approach for glaucoma detection in retinal fundus images, consisting of two steps. In the first step, a faster region proposal neural network (RCNN) was utilized to detect the optical disc (OD). In the second step, a regression network was trained to estimate the cup-to-disc ratio (CDR) by analyzing the region surrounding the detected OD. The proposed method was evaluated using the MESSIDOR and Magrabi datasets. Experimental results demonstrated the effectiveness of the approach in glaucoma detection.

Mamta Juneja et al. [22] introduced GC-NET, a deep learning-based glaucoma classification network, designed to classify retinal images as either glaucomatous or non-glaucomatous. The proposed GC-NET is evaluated on the RIM-One and Drishti datasets, achieving impressive results. The experimental outcomes demonstrated an accuracy of 97.51%, sensitivity of 98.78%, and specificity of 96.20%. The network obtained a true positive rate (Tp), a false positive rate (Fp), a true negative rate (Tn), and a false negative rate (Fn) of 0.81, 0.76, and 0.01 respectively. This performance outperformed cutting-edge techniques, demonstrating the promise of the suggested strategy for the preliminary screening of glaucoma patients.

Rahul Krishnan et al. [23] performed glaucoma detection using an existing pipeline that involved segmenting the optic disc and cup, calculating the Cup-to-Disc Ratio (CDR), and making predictions based on the CDR value. A threshold-based algorithm was used for optic disc segmentation, while a modified region growing algorithm was applied for cup segmentation and additionally blood vessel infilling and morphological operations were included. Cup segmentation was a challenging problem, and the proposed method presented a novel approach to address this challenge. The results demonstrated that the proposed approach achieves accurate glaucoma prediction with lower computational requirements.

Syna Sreng et al. [24] introduced a two-stage glaucoma screening system to alleviate the workload of ophthalmologists. The first stage involved optic disc region segmentation using a modified DeepLabv3+ architecture with multiple deep convolutional neural networks. For the classification stage, transfer learning, learning feature descriptors using a support vector machine, and ensemble methods combining both were used. Evaluation of five datasets consisting of 2787 retinal images revealed that a combination of DeepLabv3+ and MobileNet performed best for optic disc segmentation and the ensemble of methods outperformed conventional methods for glaucoma classification with high accuracy and Area Under Curve (AUC) values.

S. Phasuk et al. [26] used a combination of classification networks and an artificial neural network (ANN) for glaucoma screening. Three public datasets (ORIGA-650, RIM-ONE R3, and DRISHTI-GS) were employed to train and evaluate the proposed network. The experimental results demonstrated that the proposed network outperformed other existing glaucoma screening algorithms, achieving an AUC (Area Under the Curve) of 0.94.

Andres Diaz-Pinto et al. [27] explored the use of five ImageNet-trained models (VGG16, VGG19, InceptionV3, ResNet50, and Xception) for automated glaucoma assessment using fundus images. The results were compared with previous works in the literature using cross-validation and cross-testing strategies through extensive validation. The Xception architecture achieved the best performance, with an average AUC of 0.9605, specificity of 0.8580, and sensitivity of 0.9346 across five public databases. Additionally, a new clinical database, ACRIMA, containing 705 labeled images was introduced, which demonstrated high specificity and sensitivity, validated on all publicly available glaucoma-labeled databases.

Juan J. Gómez-Valverde et al. [28] investigated the impact of various Convolutional Neural Network (CNN) approaches on glaucoma detection. They compared CNN performance to human evaluators and examined the integration of clinical history data with images. The best result was achieved using VGG19 in a transfer learning setup, with an AUC of 0.94 and comparable sensitivity and specificity to human evaluators. Results from three datasets consisting of 2313 images indicated that this approach could be a valuable tool for large-scale glaucoma screening programs.

Jin Mo Ahn et al. [29] developed a deep learning model to diagnose glaucoma both advanced and early glaucoma using fundus photography, where logistic classification and convolutional neural network models were constructed using Tensorflow, also a trained GoogleNet Inception v3 model was fine-tuned using the same datasets. The CNN model outperformed the logistic classification model, achieving higher accuracies and AUROC values on all datasets. The transfer-learned GoogleNet Inception v3 model achieved the highest accuracies and AUROC values, indicating its effectiveness in the classification task.

U Raghavendra et al. [30] introduced a novel computer-aided detection (CAD) tool for accurate glaucoma detection using deep learning. Trained a deep convolutional neural network (CNN) with eighteen layers to extract robust features from digital fundus images, which are then classified as normal or glaucoma during testing. The proposed system achieved a high accuracy of 98.13% on a dataset of 1426 fundus images, consisting of 589 normal and 837 glaucoma cases. Results demonstrated the system's robustness, making it a valuable supplementary tool for clinicians.

3. METHODOLOGY

3.1 DATASET

The Brazil Glaucoma Database (BrG) contains labeled images for glaucoma detection studies using Deep Learning.
The database contains images of 1000 volunteers, 500 with glaucoma, and 500 without glaucoma.
It contained both the left and right eye

3.2TRANSFER LEARNING WITH UNET

Transfer learning is a powerful technique in deep learning that leverages knowledge from pre-trained models on large datasets and applies it to new tasks.
In glaucoma detection, transfer learning enables the utilization of learned features from diverse medical image datasets to identify glaucomatous features.
The U-Net model, specifically designed for medical image segmentation, is well-suited for glaucoma detection as it accurately segments key structures in retinal images.
By incorporating transfer learning into U-Net, re-trained models on large-scale image datasets enhance the segmentation capabilities and accuracy of glaucoma detection.

V-UNET V1.1

Training condition:

- Pre-train with 2000 eye datasets (train-test-split of 0.8)
    - Learning rate 1e-5, trained with 100 epochs
- Fine-tune with the same eye dataset, with 5-fold cross-validation
    - Learning rate 5e-6, trained with 100 epochs

DATA PREPROCESS: DATA AUGMENTATION

Create a double number of image and mask copies using a horizontal flip

- The function horizontal_flip to create copies
- The function gen_filepath_list to return a list of file paths of generated images and masks

IMAGEDATAGENERATOR USED FOR DATA AUGMENTATION

- function train_generator
- can generate images and masks at the same time using the same seed for the image and masks generators to ensure the same transformation between corresponding images and masks
- to be used in the model training process

TRANSFER LEARNING WITH U-NET

- pre-trained model (VGG16) for transfer learning
- function TL_unet_model(input_shape), which has VGG16 encoder frozen
- output_shape = (256, 256, 1)

TRAINING THE TL U-NET MODEL

- Splitting the dataset: 80% for training, 20% for validation and testing

FINE-TUNING

- Unfreeze the contracting path
- Apply KFold cross-validation (k=5)
- Evaluation metrics: dice score, iou
- For unfreezing use function finetune_unfreezeall
- Take the base model (pre-trained one) and unfreeze all the layers

3.3CNN MODEL

A CNN (Convolutional Neural Network) is a type of deep learning model that is commonly used for image classification, object detection, and other computer vision tasks. It is designed to automatically learn hierarchical representations of data directly from raw inputs, such as images.
CNNs are composed of multiple layers, including convolutional layers, pooling layers, and fully connected layers. Here is a brief overview of these components:

- Convolutional Layers: These layers perform convolution operations on the input data using a set of learnable filters or kernels. Each filter slides across the input, computing dot products and producing feature maps. The filters capture spatial patterns and local features at different scales.
- Pooling Layers: Pooling layers reduce the spatial dimensions of the feature maps obtained from the convolutional layers. They downsample the feature maps by taking the maximum value or average value within small, non-overlapping regions. This helps to decrease the computational complexity and make the model more robust to small spatial variations.
- Fully Connected Layers: These layers are typically located at the end of the CNN architecture. They connect every neuron from the previous layer to every neuron in the current layer, allowing the model to learn high-level representations and make predictions based on the learned features. The output layer of a CNN is often a softmax layer for multi-class classification.

- Activation Functions: Activation functions introduce non-linearity into the CNN model, enabling it to learn complex relationships between inputs and outputs. The activation function used in CNNs is ReLU (Rectified Linear Unit)

During training, CNN models learn the optimal values of the filter weights through a process called backpropagation. This involves calculating the gradient of the loss function concerning the model's parameters and updating the weights using optimization algorithms such as stochastic gradient descent (SGD) or Adam.

4. PROPOSED SYSTEM

Deep learning, specifically the combination of transfer learning with the U-Net model and convolutional neural networks (CNNs), offers promise in improving glaucoma detection accuracy.
Transfer learning leverages knowledge from pre-trained models, reducing training time and improving generalization.
The U-Net model excels at segmenting key structures in retinal images, while CNNs extract discriminative features associated with glaucomatous changes.

UNet

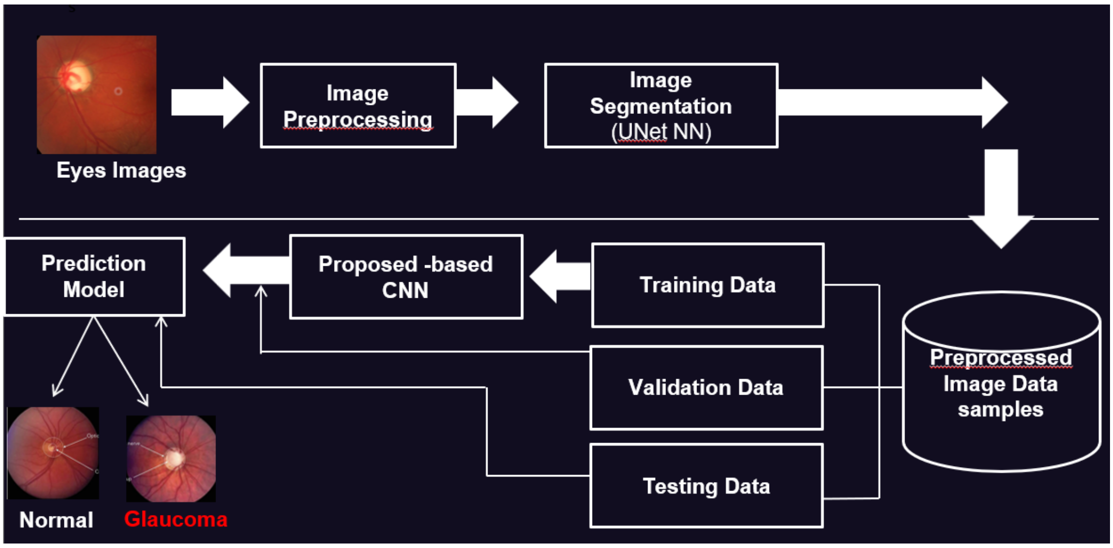

Figure 1. architecture diagram-workflow

5. RESULT AND DISCUSSION

5.1 UNET MODEL TRAINING RESULTS

```
Accuracy:  [0.9913586378097534]
Loss:  [-0.8903865814208984]
Dice coefficient:  [0.890386700630188]
IOU:  [0.8027355074882507]
```

Figure 2. unet model training results

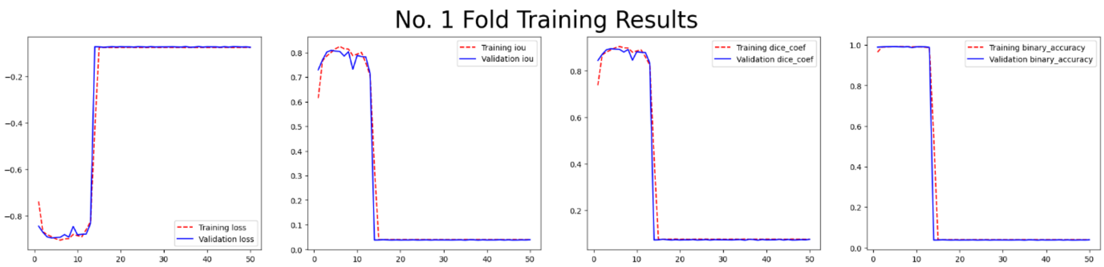

Figure 3. fold training results

5.2 SEGMENTED IMAGE

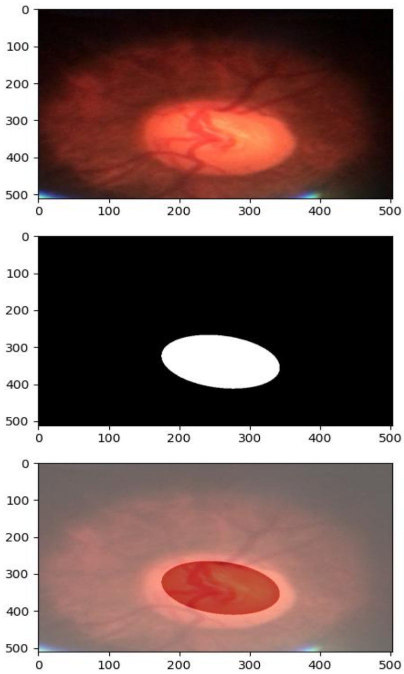

Figure 4. segmented image

5.3 CNN MODEL

From the figure, we can observe many troughs and depths in the plotted graph. Since we are having 150 epochs, the accuracy at each epoch is plotted in the graph and we can see that after 100 epochs there is a sudden drop in accuracy and when we increase the epochs there, we can see a steady increase in the model's accuracy. This will give us a better insight into the model's performance.

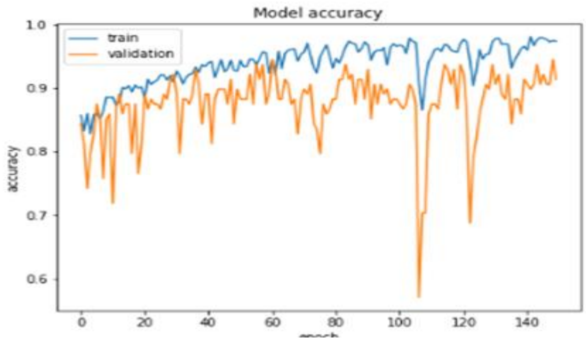

Figure 5. model accuracy

The figure depicts the different values of loss at each epoch. As we have discussed earlier, there is a sudden drop in the accuracy as the loss function value is varied. This graph of loss function helps us to predict the issues that occur with learning. These issues can lead to underfitting or an overfitted model.

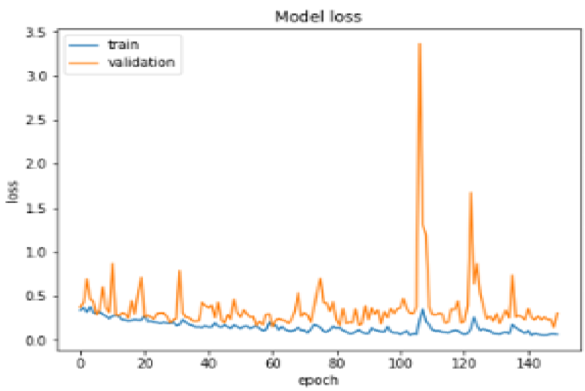

Figure 6. model loss

5.4 CLASSIFICATION REPORT

The classification report here instigates the fact that this model which achieved an accuracy of 98.47% with an integrated dataset is to be noted and this is the first-ever model to achieve this. All the existing studies have achieved the results by working on small datasets whereas the proposed model has a large dataset. The CNN here can process even on low-quality images. The dataset created is a balanced dataset with a proportionate number of glaucoma and normal images.

	precision	recall	f1-score	support
glaucoma	0.984615	0.984615	0.984615	65.000000
normal	0.984848	0.984848	0.984848	66.000000
accuracy	0.984733	0.984733	0.984733	0.984733
macro avg	0.984732	0.984732	0.984732	131.000000
weighted avg	0.984733	0.984733	0.984733	131.000000

Table 1. classification report

6. CONCLUDING REMARKS AND FUTURE ENHANCEMENT

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